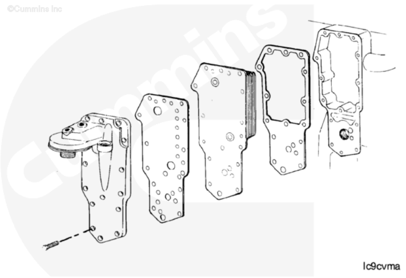
Trainings

General Information

The lubricating oil cooler is mounted between the lubricating oil cooler cover and the cylinder block. Since neither the lubricating oil cooler or lubricating oil cooler cover can be removed without removing and installing the other, this procedure covers the removal and installation of both components.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

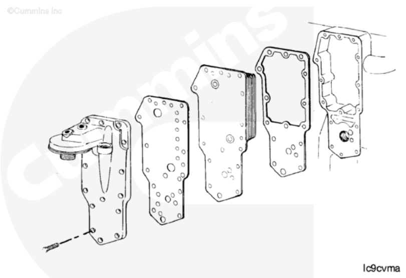
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html)
- Clean around the lubricating oil cooler cover
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/40/40-007-013-tr.html)
- If equipped, disconnect the turbocharger oil supply line. Refer to Procedure 010-046 (/qs3/pubsys2/xml/en/procedures/40/40-010-046-tr.html) in Section 10.

Remove

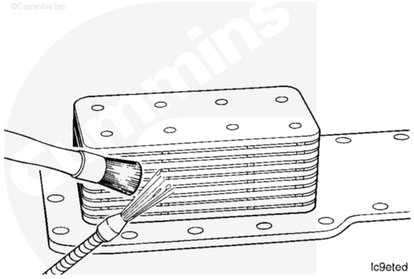
Remove the lubricating oil cooler housing capscrews, housing, gaskets, and cooler element.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

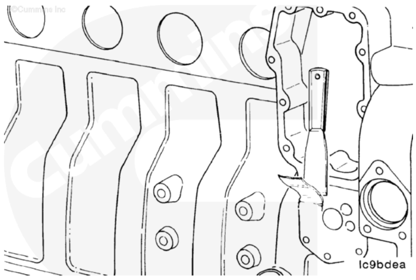
Use a solvent that will not harm copper to clean the oil cooler elements.

Use solvent to clean the oil cooler housing and cover.

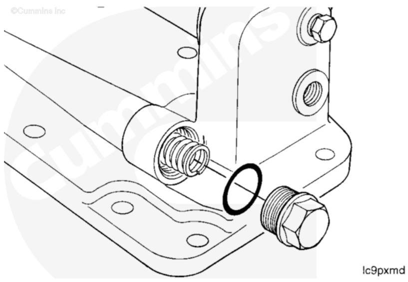
Note : Replace the lubricating oil cooler if any debris is found or the engine has had a debris-causing failure.

Clean the sealing surfaces.

When cleaning the oil cooler cover, be sure to clean the lubricating oil bypass valve.

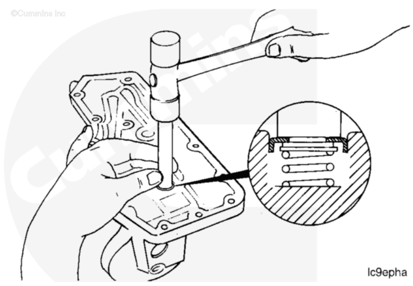


If any debris is suspected to have gone through the engine or troubleshooting a lubricating oil pressure issue, remove and inspect the lubricating oil pressure regulator located in the lubricating oil cooler cover. Replace if necessary. Refer to Procedure 007-029 in Section 7 (/qs3/pubsys2/xml/en/procedures/40/40-007-029-tr.html).



If any debris is suspected to have gone through the engine, inspect oil filter bypass valve located in the lubricating cooler cover. Make sure the valve is fully seated and opens and closes freely. Replace if necessary.

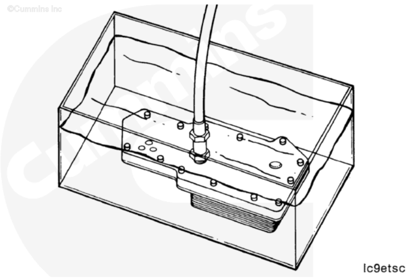
Note : The bypass valve requires a 345 kPa [50 psi] pressure differential to open.



Leak Test

Use the lubricating oil cooler pressure test kit, Part Number 3823876, to pressure-test the element to check for leaks. If leaks are detected, replace the element.

Air Pressure Test		
kpa		psi
449	MIN	65
518	MAX	75



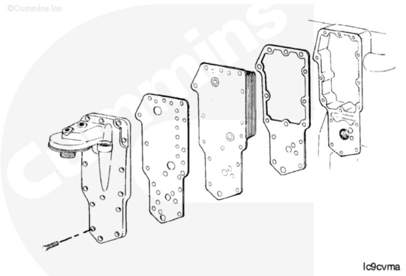
Install

Note : Be sure to remove the shipping plugs from the oil cooler element.

Note : When installing a new lubricating oil cooler be sure to use the correct part number. Replace with the same part number or use the engine serial number and QuickServe™ OnLine to ensure the use of the correct part.

Note : To make sure of the use of compatible gasket combinations, it is essential that the technicial confirms both the oil cooler and filter head gaskets correctly mate with all fluid passages on the cylinder block, oil cooler element (both sides), and the filter head surfaces. This must be done by placing the gaskets on each of these surfaces, in turn, to confirm compatibility prior to final assembly.

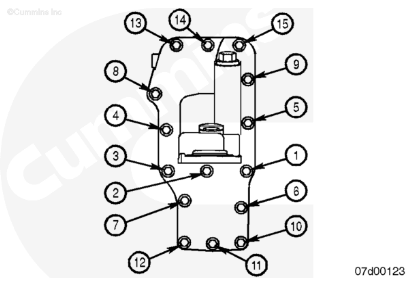
Assemble the lubricating oil cooler cover, capscrews, gaskets, and oil cooler.



If the engine uses a lubricating oil cooler cover in which the lubricating oil filter is mounted low, use the torque sequence shown.

Note : Snug capscrew numbers six and eight, then tighten in the sequence shown.

First Stage Torque Value	17 Nm	150 in-lb

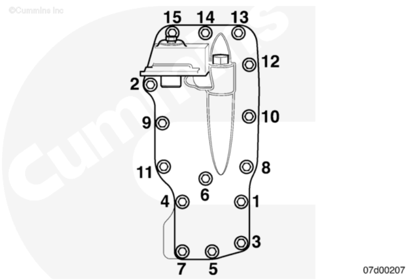


Second Stage Torque Value	28 Nm	248 in-lb

If the engine uses a lubricating oil cooler cover in which the lubricating oil filter is mounted high, use the torque sequence shown.

Note : Snug capscrew numbers six and eight, then tighten in the sequence shown.

First Stage Torque Value	17 Nm	150 in-lb
Second Stage Torque Value	28 Nm	248 in-lb



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

If the engine does not produce oil pressure in 15 seconds after starting the engine, shut off the engine to avoid component damage.

- If equipped, connect the turbocharger lubricating oil supply line. Refer to Procedure 010-046 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-046-tr.html)
- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/40/40-007-013-tr.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)
- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/40/40-008-018-

tr.html) in Section 8.

- Connect the batteries
- Operate the engine and check for leaks
- Stop the engine, and check the coolant and lubricating oil level.

Last Modified: 10-Jun-2009